Design, Testing and Maintenance

Software Engineering - Units 3-4 - Quality assurance

Good design

- High cohesion: a module does one well-defined job.
- Low coupling: modules depend on each other as little as possible.
- Modularity, information hiding and clear interfaces make change cheap.

Testing levels

- Unit testing - one function or class in isolation.
- Integration testing - modules working together.
- System testing - the whole product against requirements.
- Acceptance testing - the customer decides it is fit for use.

Black-box vs white-box

Black-box techniques ignore the code: equivalence partitioning and boundary value analysis. White-box techniques use the code structure: statement, branch and path coverage.

> *Boundary value example: an age field accepts 18 to 60. Test 17, 18, 19, 59, 60 and 61.*

Cyclomatic complexity

```
V(G) = E - N + 2P
  E = edges, N = nodes, P = connected components
Shortcut: V(G) = number of decision points + 1
```

V(G) gives the number of independent paths, and so the minimum number of test cases needed for full branch coverage.

Maintenance and quality

- Types: corrective, adaptive, perfective, preventive.
- Version control (Git), code reviews and continuous integration keep quality high.
- Technical debt is the future cost of shortcuts taken now.

NoteNest demo material - self-created for illustration, not an official publication.